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Two Three Step Authentication in ATM Machine to Transfer Money and for Voting Application

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Abstract

Voting plays a major role in electing a right person by the public to rule the country. The main aim of the paper is to perform two operations such as transaction of money and for voting application through ATM machine, by providing the authentication like Biometric – Fingerprint and Face Recognition through the comparison with the Aadhar car for more security and privacy. This voting application through ATM's makes more easier and faster for the people to increase the percentages of votes.

Keywords: Automated Teller Machine (ATM), Personal Identification Number (Pin), One Time Password (OTP), Short Message Service (SMS).

1. Introduction

ONLINE money transaction can be done through ATM machine. The authorized person goes to nearby ATM's in order to transact money easily. In the ancient days people withdraw or deposit money into their account only through the respective banks. Nowadays due to the development of technology we are using ATM machine to transfer the

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money in account. It is very easier for the people to withdraw or deposit money from anywhere, anytime via the ATM cards and pin number. But this process is not more secure because any candidate can transfer money if ATM card and pin number has been known. Thus the amount can be theft by anyone without the account holders knowledge. In this paper, we are using (one time password or pin) OTP. It is more security. The authorized person PIN number verification is completed then the device will be generate the One Time Password and convey to registered mobile in the database. Once OTP number is correctly entered in the ATM machine, money transaction takes place.

Online voting scheme is feasible to increase the percentage of voting system. In traditional voting system paper based voting is followed, so we propose ATM voting which replaces paper as a digitized one. Nowadays most of the voters are very busy in his/her works and most of the voters are living far away from voting centre. Some voters hesitate to wait for the queue in election booth which leads to decrease in percentage of voting system. As over crowd in the electing booth, old people avoid to vote for the candidate due to health issues. These are the some main disadvantages of traditional voting system. In this paper, we introduce a newest voting method based on ATM. ATM machine in all the states and all the countries are in large number of availability. People can vote through nearby ATM machine. So, percentage of voting will get increase. Authorized person voter id verification and Aadhar card number verification is used in the proposed system in order to increase the security. In addition to that biometric facilities such as Fingerprint and Face Recognition are implemented. As a results number of voting is increased in a secure and easy manner.

1.1. Existing Methodologies

In the earlier days people withdraw or deposit money into their account only through the relevant banks. In order to wait for queue withdraw or deposit money in our bank account. So, waste of time. In traditional voting system paper based voting is followed, so we propose ATM voting which replaces paper as a digitized one. Some voters hesitate to wait for the queue in election booth which leads to decrease in percentage of voting system. As over crowd in the electing booth, old people avoid to vote for the candidate due to health problem.

1.2. Origin of Proposed System

The result of the proposed system is to transfer the amount with more security. Thus our money in our account cannot be theft by anyone since we are using two safety measures such as PIN and OTP generation to the registered mobile with the bank account. The second important impact of this system is we can vote the candidate in the election through ATM machine itself through the four step process. They are:
1.Enter a Voter Id Number
2.Aadhar Card Number
3.Biometric(FingerPrint and Face Recognition) which will cross check with the already registered in Aadhar Card. If all the first three process has been verified successfully then the fourth process will be the Voting process in which we can vote the candidate by selecting the authenticated party.

2. Related Work

Mithun Dutta, et al[1]To guarantee the security at the same time as undertaking transaction via swipe device, the user will verify the transaction by using an approval message via GSM technology. The bio-metric features and SMS authentication technology to advance the security. Here, fingerprint identification is verified together with PIN verification to recognize the real user for the period of an ATM transaction. It is not possible to duplicate the biometric features like fingerprint, to resolve the verification trouble, this concept can be a pleasant response. All through the buying or fee via the swipe system, the user will get an affirming SMS via GSM technology in his/her cellular wide variety which is registered with the bank to approve the transaction. To see the illegal/fraud, user in both cases, the region can also be traced the use of a GPS modem for the during the time of transaction. This will assist to reduce the opportunity of kidnap and redouble the safety of ATM and swipe device. It occurs an problem, approving message then to verify that he/she is the right card holder some registered records will be necessary to verify such as date-of-birth, mobile number, address, blood group, signature and etc. This verification will not permit the user to make purchases but it'll verify whether the is a prison card holder or not. If the card holder doesn't save the accurate records which is stored in have an account database, then the system will automatically become aware of him/her as illegal user and the card can be blocked at once. The system will notify the real card holder/person about the incident by sending the message. It is a good way of message exchange via GSM.

Apurva Taralekar, et al[2]There are various techniques used for biometric technology, such as Iris Technology, facial recognition, voice recognition, fingerprint technology, palm technology, signature recognition, etc. In this paper used for fingerprint biometric technology/device as high accuracy, low price as compared to other technique or device. Each user has several bank accounts in different banks, persons want to carry several ATM cards for transactions, there may be unique PINs for every account. At the times it's occurring that we forget ours PINs, lose our cards, cards get stolen, stolen PINs such scenario are facing in our daily life, so to overcome this problem, "Layout Single Touch Multi-banking Transaction System using biometric and GSM authentication" So, it is simple to use. The issue like fraud, illegal access, cards getting stolen forgetting the PINs are prohibited. So, ATM card contains the number of disadvantages like breaking card, losing card, stolen card, losing PIN, forgot PINs, etc.

Prakash Chandra Mondal, et al[3]Three factor authentication using online writing signature verification from user account profile database conjunction with chip based totally card and PIN after considering cost involvement, effectiveness and system possibility. The lot of methods based on two or three factor verification for ATM transaction have been evolve in the last few decades. A variety of technology and methodology have been exercise by the financial institution in order to ensure at most security for the customers. ATM offers credit cards, fake withdrawal of cash through the use of fake or stolen bank cards has emerged as a critical problem. Many signatures matching algorithms have already adopt which are found to be valuable and working in the real world. Today, ATM is underneath hazard by way of fraudsters like card skimming, cash trapping, fake assistance, shoulder surfing, eavesdropping, fake PIN pad overlay, keep-use etc.

Nischal Bansal, et al[4]The patron needs for money, have a login to their cellular banking and cash withdraw transaction like an ATM system the use of "cash withdrawal" characteristic. In few years back, money withdrawals in your bank account, we have to visit a bank and waiting for long queue and fill the form, after that we get the money. So, we have spent a lot of times. After that, new technology move forward to the ATM machine by using swipe a smart card at anytime, anywhere, anyplace etc. Little minutes get the cash in your account without fill a form. But still sometimes, we have to wait for the queue at the ATM machine. ATM device has money withdrawal, in much less time growing level of safety.

Bharathiraja N, et al[5]HAAR is a method that is used to recognize the face area in an image. A window is moved over the given image to detect only the face area. The input picture is split into pics, namely high-quality photo (picture with face) and negative picture (image without face). Having a greater quantity of high-quality and poor pics will normally purpose a extra accurate classifier. The most important issue includes shoulder-surfing attacks, replay attacks, card cloning, and PIN sharing. Serveral researches have additionally has been performed to make structures, helping card-much fewer transactions. User go to login software via the use of username and password. The duration of login the IMEI wide variety of the registered cellular is automatically detected. Username, password in addition to an IMEI wide variety of user mobile are used right here to authenticate user. The app permits a user to experiment a QR code from the display screen of an ATM terminal and connects to the financial institution's server to attain comfortable one-time-password (OTP). The drawback of the machine is authentication system will take a great deal time. It's miles a time consuming process.

Aman Kumar [6]This technology work in three main steps or stages, in first stage user need to swap the ATM and used a palm for scanning after effectively swap device will generate voice you can enter for 2nd stage and palm will save in that transaction database. The second phase have to scan for retina and third stages will final ATM. Generate as many troubles for the duration of theft of the ATM like police not attain incorrect time, no data or no alert, weak wall of ATM room. The CCTV cameras monitored by police station. Any thief does theft in ATM, then, alarm generates in police station and gasoline will leak in nearest ATM. Then in this situation all data like, theft time, date, CCTV records, voice records will saved in the server. The server monitored by means of country level. If any condition the internet lost, then we cannot send the communication.
P. Umadevi, et al[7]To avoid phishing emails a mathematical system education technology is used to filter out the possible phishing emails; But, the sort of content filter out doesn't paintings rightly all of the time. Blacklists of phishing mail servers are constructed in; but, those servers are not useful when an attacker hijack a epidemic infected system. In a route primarily based verification has been introduced. Irrespective of the capability challenge considered in an execution, such password hashing technology has a roaming issues and other issue. The virtual password is a dynamic password. It is used for one of the authentication. It is a small amount of individual computing and safety password or PIN for the online environment. User ID, password stolen to the attacker.

Sufian Hameed, et al[8]The Safe Pass new panic password that may be without difficulty deploy over the ATM framework. In Safe Pass, unique attention is given from the usability attitude because it could be extraordinarily valuable for the large scale increase. ATM terminal is prompt fake money also issued. Panic passwords are used for effective indicator and then pressure through verification. In duress, state of interaction the user simplest desires to exchange the very last digit of the PIN. This may cause the duress, state of affairs and the ATM system will reply in accordance.

H. Lasisi, et al[9]The character has been enrolled in a organization, he/she can begin on to use biometric technology to have access to person monetary records to authorized transactions. The authorize device, software validate the user first of all with the pin code before the utility activate the sensor and fingerprint for verification. A user who wants to fingerprint has already has been recorded may not be known. This is called "replica rejection" and happens to any technology and manufacturer. More than one security compliments are compulsory to provide an excessive level of safety against fraud and different threats. fake rejections happen because of a error during registration, with the capture of a biased thumb print, usually the tip or surface.

X.Ignatius Selvarani, et al[10]The voting system is used for smart mobile application via the internet. Like vote for online registration, voter to cast vice, and then display the results. It is more efficient, vote for voters and then vote anywhere, anytime, anyplace. It's security provides through RSA encryption algorithm used. It is more time and cost consuming and also affects the user from long distance to cast their vote from a remote place. Only vote for authorized person to access a login. The smart mobile phone is used to only limited resources, time efficient, helpful in the remote voting process. Once you can vote for authorized person no one can change with the application.

3. Proposed Method

Two features of proposed methods are banking and voting scheme via automatic teller machine. In banking system, admin section can be already authorized person details are stored in the database. If the PIN number verification is successfully, then the OTP generated to the registered mobile phone. Correctly entered in the ATM machine, OTP verification successfully, it is more security. Next, money can be transaction in your account.

In voting system, we can vote the candidate in the election via ATM machine itself. Authorized person voter id verification and Aadhar card number verification is used in the proposed system in order to increase the security. In addition to that biometric facilities such as Fingerprint and Face Recognition are implemented as shown in the Fig1. As a results number of voting is increased in a secure and easy way.

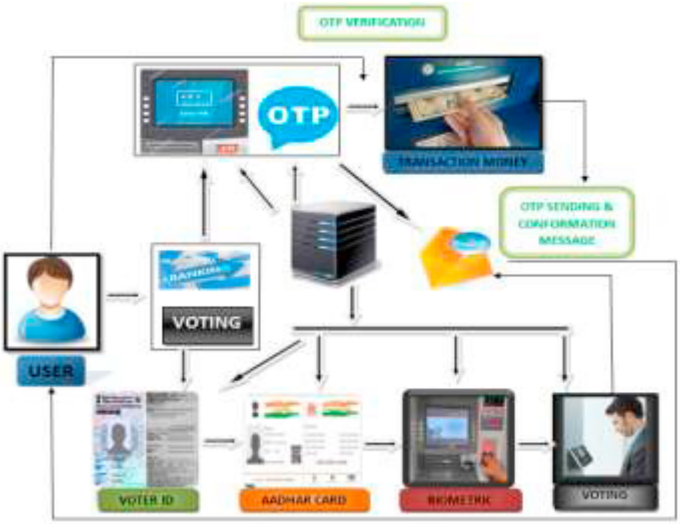

Fig 1: Proposed System Architecture

Implementation of proposed system is divided into two major modules. They are:
1. Banking
2. Voting

3.1. Banking

In the ancient days people withdraw or deposit the amount into their account only through the respective banks. Nowadays due to the development of technology we are using ATM's to transfer the amount to our account. This makes very easier for the people to deposit or withdraw money from anywhere and at anytime through the ATM cards and pin number. But this process is not more secure because any candidate can transfer money if ATM card and pin number has been known. Thus the amount from the amount can be theft by anyone without the account holders knowledge. So we introduced the three step process to transfer the amount from bank account through ATM's. They are:

(i) PIN number verification.
(ii) OTP Generate to the registered mobile number.
(iii) Transaction of Money.

3.2. PIN Number Verification

All our details has been saved as an database in the admin section of the bank. Thus when we insert an ATM card into the machine first it will ask to enter the pin number which has been generated by the user during their registration of the card. When the pin number has been entered the machine will cross verify the pin number of the particular card with the already saved database. If the verification is done successfully then the machine moves on into the second step.

3.3. OTP Generation

Once the pin verification has been completed then the device will be generate the OTP (One Time Password) to the registered portable number in the database. If the OTP has entered is correct then only the machine moves on to the step 3 else the machine will end up the transaction process by rejecting the ATM card. So this OTP generation process plays an important role in the form of security. If someone lost their ATM card along with the PIN number there is no possibility of thefting of money from the card holders account because the OTP has to be correctly enter.

3.4. Transaction

If all the above two steps has been verified successfully then only the person can able to take the cash from the account. The transaction part will ask the amount to be withdraw from the account and all other process in the transaction section are same as in the current system.

3.5. Voting

Voting is the basic rights for all the people in the country. But the major disadvantage of the Voting system is 100% voting is not possible by the people due to various reason such as,

(i) Over crowd in the election booth so most of the old people will avoid to vote the candidate.

(ii) Some people cannot able to come to particular voting booth to which allocating for them due to survival in other states etc.

The ATM's plays vital role in all the states and all the countries and large number of ATM's are also available around the world. Due to this we have introduced the voting system in the ATM's itself to avoid the above problems.

This can be done by the following four steps:

(i) Voter id verification.
(ii) Aadhar card verification.
(iii) Biometric (FingerPrint and Face Recognition).
(iv) Voting process.

This Voting module will be enable only on the time of the Election.

(I) Voter ID Verification

The Voting module has the first step of voter id verification process. Each person in the country will have the voter id card and the reference id will be unique for everyone and all the information of the particular person will be stored as an database by the Election Commission. Voter id card is the basic and important identification to participate in the election and have the rights to elect the candidate. So the reference id in the voter id card should be enter into the ATM and it must be cross check with the already stored database. So we can able to vote anywhere using ATM with the voter id card.

(II) Aadhar Card Verification

The second step of the Voting module is the Aadhar Card Verification. In this we have to enter the Aadhar Card Number in the ATM. This process is used in Biometric section for more security. Because the fingerprint for all our ten fingers and the detection of our face has been captured and stored as the database for every individual persons during the registration of the Aadhar Card.

(III)Biometric

The Biometric (Fingerprint and Face Recognition) is used in this process for more security. Because the voting system should be secured to avoid the misuse of the Voting system in ATM's.

(IV) Voting Process

After all the above section has been verified successfully the machine enters into the voting section. In which it displays all the election candidates for the current election which has been updated by the Election Commission. The Buttons in the ATM is used to select the particulate candidate by the person

4. Result and Discussion

Online transaction money from your account via ATM machine, Personal Identification Number(PIN) verification, One-Time Password(OTP) generated, Transaction Money(TM). Online voting system via ATM machine, Voter ID card verification, Aadhar card verification, Biometric(Fingerprint, Face Recognition), and Voting process.

4.1 Online Money Transaction

Online money transaction in ATM machine, authenticated person should swipe a card and then in PIN entry module, type 4-digit of your PIN number for verification . The OTP will be generated to the registered mobile number. Check your mobile phone for OTP verification number. Then, type your OTP verification number in ATM machine so that money will be transacted. By this method transaction process will be more secure.

4.2 Online Voting System

Online Voting system in ATM machine, Voter ID card and Aadhar card number should entered into the ATM machine, if it is verification of your number. By providing the authentication like Biometric – Fingerprint and Face Recognition. By this method voting process will be more secure. To avoid misuse of the voting system in ATM's. ATM machine enters into to voting section, displays all the election candidates for the current election has been updated for Election commission. Authorized person should be voted. This Voting system in ATM's makes easier and faster for the people to increase the percentages of votes.

5. Conclusion

The Conclusion of the Banking and Voting module is to transfer the money from the account with more security of PIN and OTP generation. The Voting module in the ATM's is used to reduce the crowd in Election Booth and most people can be vote the candidate during election time. Thus the percentage of the voting will be increase and also it is more secure due to verification of the Biometrics (FingerPrint and Face Recognition).

References

[1] Mithun Dutta, Kangkhita Kaem Psyche, Tania Khatun, Md. Ashiqul Islam, Md. Azijul Islam, "ATM Card Security Using Bio-metric and Message Authentication Technology", IEEE International Conference on Computer and Communication Engineering Technology (CCET), 2018, PP.280-285.

[2] Apurva Taralekar, Gopalsingh Chouhan, Rutuja Tangade, Nikhilkumar Shardoor, "One Touch Multi-banking Transaction ATM System using Biometric and GSM Authentication", International Conference on Big Data, IoT and Data Science (BID), 2017, PP.60-64.

[3] Prakash Chandra Mondal, Rupam Deb, Md. Nasim Adnan, "On Reinforcing Automatic Teller Machine (ATM) Transaction Authentication Security Process by Imposing Behavioral Biometrics", Proceedings of the 2017 4th International Conference on Advances in Electrical Engineering (ICAEE), IEEE, 2017, PP.369-372.

[4] Nischal Bansal, Nepali Singla, "Cash Withdrawal from ATM machine using Mobile Banking", International Conference on Computational Techniques in Information and Communication Technologies (ICCTICT), IEEE, 2016.

[5] Bharathiraja N, Ravindhar N.V, Loganathan V, "Secure PIN Authentication for ATM Transaction using Mobile Application", International Journal of Soft Computing and Engineering (IJSCE), 2016, PP.60-63.

[6] Aman Kumar, "Advance Security System for ATM", International Journal of Scientific Research Engineering & Technology (IJSRET), 2015.

[7] P. Umadevi, V.Saranya, "Stronger Authentication for Password using Virtual Password and Secret Little Functions", ICICES, IEEE, 2014.

[8] Sufian Hameed, Syed Atyab Hussain, Sohail Hussain Ali, "SafePass: Authentication under Duress for ATM Transactions", 2nd National Conference on Information Assurance (NCIA), IEEE, 2013,PP.1-5.

[9] H. Lasisi, A.A. Ajisafe, "Development of Stripe Biometric Based Fingerprint Authentications Systems in Automated Teller Machines", 2nd International Conference on Advances in Computational Tools for Engineering Applications (ACTEA), IEEE, 2012,PP.172-175.

[10] X.Ignatius Selvarani, Shruthi.M, Geethanjali.R, Syamala.R, Pavithra.S, "Secure voting system through SMS and using smart phone Application, IEEE.